

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mining Engineering) SEMESTER - 3 Winter 2025 (Supply.)

Course :Bachelor of Technology (Mining Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTMIC301 - INTRODUCTION TO MINING ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 **The process of excavating a shaft is called:**
a. Sinking b. Raising c. Drifting d. Stoping
- 2 **Freezing method in shaft sinking is used when:**
a. Rocks are too hard b. Groundwater is excessive and soil is loose
c. Shaft depth is small d. No lining is required
- 3 **A raise is driven:**
a. Downward from a lower level b. Upward from a lower level
c. Horizontally along strike d. Inclined from surface
- 4 **The first step in the mining process is**
a. Drilling b. Prospecting c. Blasting d. Transportation
- 5 **Sampling in mining means:**
a. Removing ore waste b. Collecting representative material for testing
c. Crushing and grinding d. Transporting ore
- 6 **Muck removal during shaft sinking is usually done by:**
a. Belt conveyor b. LHDs only c. Buckets or skips d. none
- 7 **The term 'stemming' in blasting refers to:**
a. Explosive charge b. Material used to fill the top of the blast
c. Detonating cord d. Drill bit type
- 8 **Ventilation in drifting is often maintained using:**
a. Natural air only b. Auxiliary fans and ducts c. Water pumps d. Lighting system
- 9 **The primary objective of mining is to:**
a. Protect the environment b. Extract minerals economically and safely
c. Increase land area d. Manufacture equipment
- 10 **The conveyor used at the coal face in longwall mining is:**
a. Belt conveyor b. Armored Face Conveyor (AFC)
c. Scraper conveyor d. Mine car conveyor
- 11 **Which of the following is an example of surface mining?**
a. Longwall mining b. Room and pillar c. Open-pit mining d. Block caving
- 12 **The percentage of coal extracted during development in bord and pillar mining is usually:**
a. 100% b. Very low (10–30%) c. 75% d. 50%

Q2. Solve the following.

- A) Write various methods of exploration. In details 6
- B) Explain unit operation (production cycle) involve in underground coal mining 6
- Q3. Solve the following.
- A) What is Deepening and Widening of shaft . 6
- B) Explain in detail the preparatory arrangements in Shaft sinking 6
- Q4. Solve Any Two of the following.
- A) Define the term: a) benches b) overburden c) stripping ratio 6
- B) How Mining contribute to civilization ,Explain in detail. Also write down Consequences of Mining . 6
- C) What are the various Mechanical methods of drivage of Roadway and Tunnels. 6
- Q5. Solve Any Two of the following.
- A) Enlist the special methods of shaft sinking . Explian anyone in details 6
- B) Explain with diagram various mode of entries in underground mining 6
- C) Explain in details drivage work in varying ground condition using conventional methods of small and medium size Tunneling and Drifting. 6
- Q6. Solve Any Two of the following.
- A) Write down the comparison between underhand stoping and overhand stoping methods 6
- B) Write down the process , Development of Mineral deposits or coal seam in UG mining with diagram 6
- C) Write down difference between surface mining and underground mining 6

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